

# Rik L.E.M. Ubaghs



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## Bio

Biomedical engineer and founder with expertise in neurotechnology, optical systems, medical devices, and data analytics. I specialize in developing and translating advanced imaging technologies into practical clinical applications, connecting cutting-edge research with meaningful medical impact.

## Education

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|------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>ETH Zurich, Zurich, Switzerland</b><br><i>Ph.D. Neurotechnology</i>                                                       | July 2017 - May 2023  |
| <b>ETH Zurich, Zurich, Switzerland</b><br><i>CAS Entrepreneurial Leadership in Technology Ventures</i>                       | Mar 2025 - Mar 2026   |
| <b>University of Amsterdam, Amsterdam, the Netherlands</b><br><i>Research Master Brain and Cognitive Sciences, Cum Laude</i> | Sept 2014 - June 2016 |
| <b>Maastricht University, Maastricht, the Netherlands</b><br><i>Bachelor Psychology and Neuroscience, Cum Laude</i>          | Sept 2011 - June 2014 |

## Skills

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|----------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>          | Dutch (Native), English (Fluent), German (Basic)                                                                                   |
| <b>Programming:</b>        | Python, Rust, Matlab, R                                                                                                            |
| <b>Databases/Querying:</b> | SQL, MySQL                                                                                                                         |
| <b>Skills:</b>             | Data Science, Biomedical Imaging, Optical Systems,<br>Optical design (Zemax, Optiland)<br>Mechanical Design (Autodesk, Solidworks) |

## Industry Positions

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| <b>Arago Labs</b><br><i>Co-founder, CEO</i>                                                                                                                                                                                                                                                                                                                                                    | June 2023 - Current |
| <ul style="list-style-type: none"><li>• Founded an early-stage medical technology venture focused on improving surgical precision in oncology.</li><li>• Designed advanced optical imaging platform combining nonlinear microscopy, and ML/DL-based tissue classification.</li><li>• Defined market entry strategy, pricing model, reimbursement pathways, and go-to-market roadmap.</li></ul> |                     |
| <b>Agora Technology, Maastricht, The Netherlands</b><br><i>Founder</i>                                                                                                                                                                                                                                                                                                                         | Nov 2016 - Feb 2018 |
| <ul style="list-style-type: none"><li>• Built data solutions that aimed at streamlining/automating the workflow of medical facilities, especially focusing on surgical equipment.</li><li>• We focused on internal logistics, and the creation, maintenance, and analysis of large-scale medical equipment databases using ML/DL.</li></ul>                                                    |                     |

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## Research Positions

**Inst. of Neuroinformatics, ETH Zurich, Zurich, Switzerland**

July 2017 – June 2023

*Ph.D. Candidate*

- Development of a MRI compatible florescence microscope (full-cycle product design).
- Design of a paradigm to combine microscopy and MR imaging methods in awake rodents.
- Design of data pipelines for multi-modal data analysis.

**Center for Neuroengineering, Duke University, Durham, USA**

Dec 2015 – Nov 2016

*Visiting Scientist*

- Design of a hybrid brain decoding strategy in Non-human Primates.
- Development of ML/DL decoding pipeline for brain decoding.

**Spinoza Center for Neuroimaging, Amsterdam, The Netherlands**

Nov 2014 - Dec 2015

*Junior Research Scientist*

- MR operator.
- Design and supervision of several large-scale research projects.

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## Extracurricular Positions

**Natural Movements, The Netherlands**

March 2007 - May 2012

*Co-Founder*

- Natural Movements is a non-profit organization that aims to engage underprivileged youth in the local community through behavioral coaching and physical education.
- Worked together with social service organisations to provide youth with any additional support they required.

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## Selected publications and patents

### Publications:

- **Ubaghs, R.L.E.M.**, ... Grewe, B.F. Simultaneous single-cell calcium imaging of neuronal population activity and brain-wide BOLD fMRI. *Nature Methods* (in press).
- Rychen, J., ... **Ubaghs, R.L.E.M.**, ... Hahnloser, R. Recordings of cetacean vocal behaviour with hydrophone arrays in Northern Norway. *bioRxiv* (2023); under review *Nature Scientific Data*. <https://whalesonic.stream>.

### Patents:

- Sainz Villalba, L., **Ubaghs, R.L.E.M.** Device and method for providing histological navigation information during in-vivo surgery. European Patent Application, Filed 2026.

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## Grants and fellowships

- ETH Zurich Pioneer Fellowship
- Swiss Cancer Foundation research grant
- Galbiati Foundation research donation

April 2025 - Current  
September 2026  
June 2025